Data Sheet

EXPERIMENT NO. 04

Title: - Simulation of Transistor Amplifier with Potential Divider Bias to Study Output Voltage and Gain for an Input Signal (using Tinkercad/proteus software circuit).

Instrument & Component Required: -

Sl. No.	Apparatus / Component	Specification	Quantity
1.	Capacitor	100 Mf	3
2.	Battery	15V	1
3.	Resistor	72. KΩ	4
4.	Transistor	-	1
5.	Ground	-	1
6.	Oscilloscope	50 ms, 1 v	1
7.	Function Generator	10Hz, 6V, Sine	1
8.	Connecting Wires	–	6
9.	Bread board	–	

Important Formulae: -

Base Voltage (VB): -

$V_B = V_{CC} \times [R_2 / (R_1 + R_2)]$

Emitter Voltage (VE): -

$V_E = V_B - V_{BE}$

- VBE = 0.7 V (for Si transistor)

Emitter Current (IE) and Collector Current (IC): -

$I_E = V_E / R_E$ and $I_C \approx I_E$

Collector-Emitter Voltage (VCE) — Theoretical: -

$V_{CE} = V_{CC} - I_C \times (R_C + R_E)$

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■ Title :- Simulation of Transistor Amplifier with Potential Divider Bias to Study Output voltage and Gain for an Input Signal (using Tinkercad/ proteus Software).

■ Theory :-

The potential divider (or voltage divider) bias Circuit is the most Widely used Configuration for Setting a Stable DC operating point (Q point) in a bipolar Junction Transistor (BJT) amplifier. Unlike Simpler biasing Schemes, its main advantage is that the operating point is Virtually independent of variations in transistor Current gain (β) and temperature fluctuations.

② Base voltage (V_B): $V_B = V_{cc} \times \frac{R_2}{R_1 + R_2}$

⊙ Emitter Voltage & Current (V_E, I_E):

$V_E = V_B - V_{BE}$

$I_E = V_E / R_E$

⊙ Collector Current & Voltage (I_C, V_{CE}):

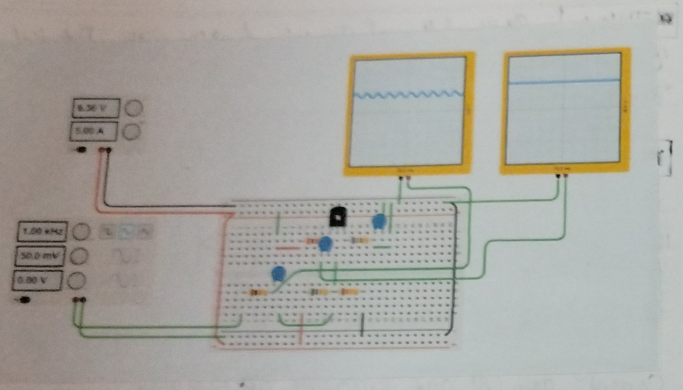
$V_{CE} = V_{cc} - I_E (R_C + R_E)$

⊙ Small - Signal AC Voltage Gain (A_V):

$A_V = V_{out} / V_{in}$

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Instrument and Components Required:-

SNO	Components	Specification	Quantity
1	Capacitor	100 μf	3
2	Battery	15v	1
3	Resistor	72 KΩ	4
4	Transistor	—	1
5	Oscilloscope	50ms 1v	1
6	Function Generator	10Hz, 6V, Sine	1
7	Connecting Wires	—	6
8	Bread board	—	1

Observation Table:-

NO	I_B	I_C	R_1	R_2	R_3	V_{CE} (practical)	V_{CE} (Theoritical)
1.	11.43	1.14	72 KΩ	8 KΩ	2 KΩ	11.9	11.91
2.	11.43	1.14	72 KΩ	8 KΩ	2 KΩ	1.88	11.91
3.	11.43	1.14	72 KΩ	8 KΩ	2 KΩ	11.92	11.91

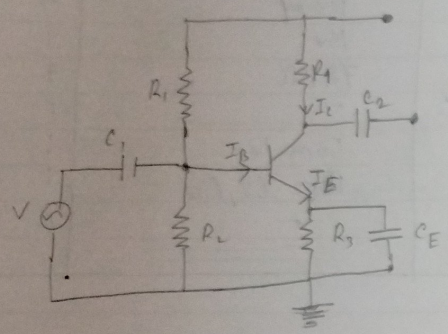
Gain Observation Table:-

NO	V_{in} (v)	V_{out} (v)	Voltage gain $A_v = V_{out}/V_{in}$
1.	20	1	50
2.	30	1.5	60
3.	40	2	50

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Circuit Diagram:-



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Calculation:-

$V_B = V_{cc} \times [R_2/(R_1+R_2)]$

$= 1.5 \times \frac{8}{2+8}$

$= 1.5V$

$V_E = V_B - V_{BE}$

$= 1.5 - 0.7$

$= 0.8V$

$I_E = V_E / R_E$

$= 0.8/670$

$= 1.193mA$

V_{CE} (Theoritical) $= V_{cc} - I_c(R_c + R_E)$

$= 1.5 - 1.193(2200+670)$

$= 11.09V$

A_V (Voltage Gain) $= V_{out}/V_{in}$

$= 100/2 \times 10^{-3}$

$= 50$

Conclusion:-

The BJT amplifier with potential divider bias was succesfuly simulated. The mesured DC collector-emmitor voltage ($V_{CE} \cong 11.09V - 11.02V$) closely aligns with the theoretical value of 11.01V. Confirming accurate operating point establishment. Furthermore the circuit effectively amplified the input AC signals achieving an average voltage gain (A_V) of approximately 50 to 60.

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